

4.13.19

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# Question

If the line  $2x + y = k$  passes through the point which divides the line segment joining the points  $1, 1$  and  $(2, 4)$  in the ratio  $3 : 2$ , then  $k$  equals

- 1  $\frac{29}{5}$
- 2 5
- 3 6
- 4  $\frac{11}{5}$

# Solution

Let the point **C** divide the line segment joining **A** and **B** in the ratio 3 : 2 where

$$\mathbf{A} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (1)$$

$$\mathbf{B} = \begin{pmatrix} 2 \\ 4 \end{pmatrix} \quad (2)$$

$$\mathbf{C} = \begin{pmatrix} x \\ y \end{pmatrix} \quad (3)$$

# Formula

By Section formula,

$$(\lambda + 1) \mathbf{C} = \lambda \mathbf{B} + \mathbf{A}. \quad (4)$$

$$\text{Here } \left( \lambda = \frac{3}{2} \right) \quad (5)$$

$$\frac{5}{2} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{3}{2} \begin{pmatrix} 2 \\ 4 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (6)$$

$$\frac{5}{2} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad (7)$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \frac{2}{5} \begin{pmatrix} 4 \\ 7 \end{pmatrix} \quad (8)$$

$$\therefore \mathbf{C} = \begin{pmatrix} \frac{8}{5} \\ \frac{14}{5} \end{pmatrix}. \quad (9)$$

# Solution

The equation of the line can be represented as:

$$\mathbf{n}^T \mathbf{x} = c_1 \quad (10)$$

$$\text{where } \mathbf{n}^T = \begin{pmatrix} 2 & 1 \end{pmatrix}, \mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} \text{ and } c_1 = k. \quad (11)$$

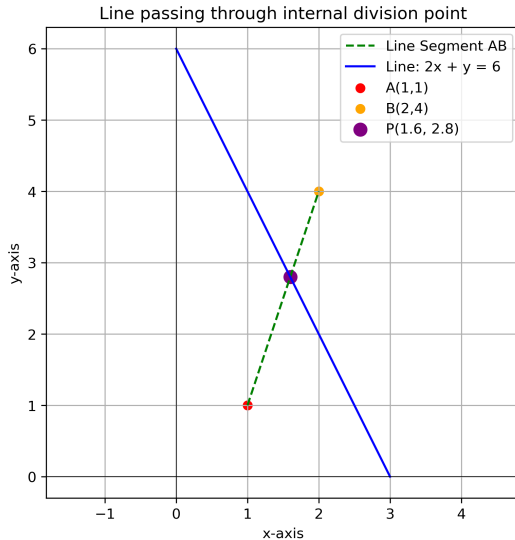
As  $\mathbf{C}$  lies on the above line,

$$\mathbf{n}^T \mathbf{C} = c_1 \quad (12)$$

$$\begin{pmatrix} 2 & 1 \end{pmatrix} \begin{pmatrix} \frac{8}{5} \\ \frac{14}{5} \end{pmatrix} = k \quad (13)$$

$$k = \frac{16}{5} + \frac{14}{5} = 6. \quad (14)$$

$$\therefore k = 6. \quad (15)$$



```
#include <stdio.h>
// sol[0] = x, sol[1] = y, sol[2] = k
void find_k(double sol[3]) {

    double x1 = 1, y1 = 1;
    double x2 = 2, y2 = 4;
    int m = 3, n = 2;

    // Internal division point (x, y)
    double x = (m*x2 + n*x1) / (double)(m+n);
    double y = (m*y2 + n*y1) / (double)(m+n);

    double k = 2*x + y;

    // Store results
    sol[0] = x;
    sol[1] = y;
    sol[2] = k;
}
```

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt

# Call C library
lib = ctypes.CDLL('./findk.so')

# Define argument types
lib.find_k.argtypes = [ctypes.POINTER(ctypes.c_double)]

# Prepare result array
sol = (ctypes.c_double * 3)()
```



```
# Call C function
lib.find_k(sol)

# Extract results
x, y, k = sol[0], sol[1], sol[2]

print(fPoint dividing AB in 3:2 = ({x}, {y}))
print(fValue of k = {k})

# Plot the graph
# Given points
A = np.array([1, 1])
B = np.array([2, 4])
P = np.array([x, y]) # Use point from C code
```

# Python Code

```
# Line equation:  $2x + y = k \Rightarrow y = k - 2x$ 
x_vals = np.linspace(0, 3, 100)
y_vals = k - 2*x_vals

plt.figure(figsize=(6,6))

# Plot line segment AB
plt.plot([A[0], B[0]], [A[1], B[1]], 'g--', label='Line Segment
AB')

# Plot line  $2x+y=k$ 
plt.plot(x_vals, y_vals, 'b', label=f'Line:  $2x + y = {k:.0f}$ ')

# Plot points
plt.scatter(*A, color='red', label='A(1,1)')
plt.scatter(*B, color='orange', label='B(2,4)')
plt.scatter(*P, color='purple', s=80, marker='o', label=f'P({x:.1
f},{y:.1f})')
```

```
# Labels
plt.xlabel('x-axis')
plt.ylabel('y-axis')
plt.title('Line passing through internal division point')
plt.axhline(0, color='black', linewidth=0.5)
plt.axvline(0, color='black', linewidth=0.5)
plt.legend()
plt.grid(True)
plt.axis(equal)

# Save the figure
plt.savefig('findk.png', dpi=300, bbox_inches='tight')
plt.show()
```